

"I Want To Be Able To Change The Speed And Size Of The Avatar": Assessing User Requirements For Animated Sign Language Translation Interfaces

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ABSTRACT

Despite research having shown that signing users vary in their preferences and needs of distinct sign language (SL) parameters, current research on SL avatars lacks consideration of the UI context's options for individualization or configuration of such parameters. Our paper addresses this gap as it presents our ability-oriented online survey, in which we asked native signers about parameters they would like to configure, as well content types that should be offered for translation within travel contexts. Our results indicate that parameters with a direct influence on the understandability of the avatar are most important for individual adjustments and situations with high time-pressure are valued highly for translation within travelling contexts. Also, our study design revealed that assessing the language background of participants in the form of an example-based self-assessment can help achieve more sensitive results, emphasizing the need for adequate research approaches.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in accessibility.**

KEYWORDS

sign language avatar design, requirements analysis, Deaf and Hard of Hearing user studies

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1 INTRODUCTION

The use of avatars for translating text into sign language (SL) is gaining momentum in Human-Computer Interaction (HCI) and accessibility research. A growing amount of studies addresses issues surrounding the appropriate design of these avatars and their SL performance, evaluating aspects such as comprehensibility, naturalness, or acceptance [14, 18, 21, 22]. However, what respective research currently lacks is the examination of the UI contexts of such avatars and with it, required functionality for adaptability and configurability regarding SL animations. Prior studies have shown that needs and preferences differ between signing users when it comes to distinct linguistic characteristics such as timing parameters [3], or in regard to design choices such as the appearance or look of an avatar [13]. From this diversity in user needs and preferences we deduct the demand for more flexibility in (avatar-based) SL translation systems, allowing users to control and adjust such parameters individually according to their own preferences and contexts. In addition, research offers little insight into differences in translation needs when considering varying types of content, i.e., are there content types that users wish to be translated more urgently than others?

This paper addresses these open research questions in a user study on requirements for the functional design of German sign language (German: "Deutsche Gebärdensprache") (DGS) interpretation interfaces in the form of a DGS-accessible, quantitative online survey with $n = 129$ native DGS signers. In this survey we asked participants to rate their personal regarding types of information that should be offered for translation, as well as parameters of the system that they would like to adjust individually - both with focus

on travel contexts. Besides prioritizations and preferences regarding varying age groups, language backgrounds, and technology affinities, our results also offer insights into conducting quantitative online studies with SL users and, specifically, the matter of (self-)assessment of language backgrounds in this target group. The main contribution of this paper thus lies in the recognition that although DGS users differ in their preferences for DGS animations as well as respective contents, there are aspects that users wish to be able to adjust by themselves, as well as different contents that they demand translations for more urgently than others - findings that should be considered and addressed as such in future research on SL translation technology.

2 RELATED WORK

2.1 Research with signing user populations

When working with or alongside SL users as a target group, previous studies have discussed a variety of subjects, such as adequate user representation [26], common missteps made by research within Deaf or Hard-of-Hearing (DHH) communities [4, 7, 23, 25], or guidelines on how to approach such a research agenda [17, 33]. In the context of missteps, issues addressed more frequently included failures to provide interpreters, insufficient explanations, or the "overwhelming focus on research questions meant to 'solve the problem of deafness'" [4]. On the matter of providing guidelines, Unger et al. [33] discussed the matter of SL accessible research in general by presenting their approach to and lessons learned from conducting ability-based focus groups with native signers. On the other side, Huenerfauth and his colleagues [17] focused on research practices for specifically evaluating SL animations with members of DHH communities, identifying best practices for collecting feedback from such target groups in respective contexts. This collection of challenges and guidelines helped us in designing and implementing our user study in a human-centered and target group appropriate manner. What they did not offer, however, were insights into how researchers can quantitatively assess SL users language background, i.e., their usage of spoken or written communication – a measure that becomes important to consider when assessing users preferences in terms of translation contents or SL individualization.

2.2 Sign Language avatars

Research investigating the use and design of avatars for translating information into SL has focused largely on evaluating specific avatars with regard to their general SL performance, i.e., their understandability, naturalness of movement or facial expression. Studies such as those from Huenerfauth, Kipp, or Heloir and their colleagues [14, 18, 21, 22] have formed a cornerstone in evaluating SL avatars by conducting focus groups with SL users, discussing which aspects of a previously designed avatar should be optimized and whether translation technologies such as these could be accepted within SL communities. Later studies as for instance from Ebling & Glauert [13] have followed this approach and also evaluated their animations through the use of focus groups with SL users. In addition to evaluation measures from [14, 18, 21, 22], Ebling and Glauert expanded their research interest on design aspects of the avatar (e.g., clothing), while others focused their entire study on

single details of the animation: While Jennings et al. analyzed technical specifications of the animation design [19], Roelofsen et al. [32] or Verlinden et al. [34] analyzed solely the comprehensibility of the displayed sign language. Specific linguistic characteristics and their consideration, influence, or implications were examined in a variety of studies [1–3, 15, 16, 29, 35] and with it, the effect of varying animation quality on user perception in [9] or the influence of participants backgrounds on their impression and acceptance of an avatar [20, 30].

All of these studies have contributed a large amount of valuable data, insights, and guidelines for the design of systems providing SL translation through animation, establishing a fundamental knowledge base on which aspects need consideration. However, they all focus on the avatar and its performance of sign language, leaving the interface in which such animations are embedded and aspects of individual options for adaptation with little to no consideration. Our paper addresses this gap in research throughout the following sections by presenting an DGS-accessible user study on requirements for the functional design of animated SL translation interfaces.

3 OUR APPROACH

The study presented in this paper was designed as part of the AVASAG research project¹ and pursued the general notion of Ability-Based Design [27, 36, 37] by understanding the target group in terms of their abilities, i.e., as sign language users. The overall goals of this study were to 1) analyze which content types are most relevant for translation into German Sign Language (DGS) within the context of travel information and services and 2) identify parameters that DGS users wish to adjust when using avatar-based SL translations. Thus, the study was conducted by means of an ability-oriented², DGS-accessible online questionnaire. Details on the study and questionnaire design, the participants as well as the results that we deduced from it are presented throughout the following sections.

3.1 Procedure & questionnaire design

Over the course of two weeks, an online questionnaire collecting requirements for an avatar-based digital translation service in the context of travel was provided to the German Deaf community via LimeSurvey [24]³. After giving their informed consent, participants provided their demographic information, including proficiency in DGS, written and spoken German. In order to contextualize the survey responses, we then administered the Affinity for Technology Interaction (ATI) Scale [6] which assesses attitude and towards and usage of technology on a four-point Likert scale (1 = strongly disagree, 4 = strongly agree). Subsequently, the survey inquired the respondents' need for translations from written German to DGS for various types of content associated with travel-related

¹The research project AVASAG [8, 28] focuses on the development of an algorithm that automatically translates German text into German Sign Language with the translation being carried out by an avatar and the initial project scope focused on the topic of travel information and services.

²When considering the ABD-approach as a whole, it includes more principles than solely a focus on abilities. It also demands for instance that systems adapt to users and not the other way around [36, 37] or that contextual en- or disabling factors are included when considering users abilities [27]. Our study, as presented in this paper, pursues mainly the first, i.e., the focus on abilities, and pays less attention to the other principles. We thus frame our study as ability-oriented, emphasizing thereby our general mindset, without claiming to fully apply all ABD principles.

³The 30-minute survey could be answered on mobile and desktop devices.

websites (e.g.: search results, announcements, news). Next, the participants were asked about their personal opinion regarding the importance of potential technical features in an avatar-based translation system, settings pertaining to the avatar's visual appearance, and functionalities of websites and applications offering such an avatar integration.

Following the ability-oriented approach, each question was posed in German written text and accompanied by an individual video translation to DGS. All response options were numbered and referenced accordingly in the video. We additionally opted for single and multiple choice response formats and refrained from including open text questions that we expected to result in a higher dropout rate among individuals with low German written language proficiency. However, one optional text box for general feedback and comments was provided on the final page of the questionnaire.

3.2 Participants

The survey link was shared via video-based social media posts in DGS with German subtitles. 177 people participated in the online survey. 45 participants failed to fill out at least half of the questionnaire and were, therefore, excluded from the analysis due to insufficient data quality. Moreover, three respondents did not meet the inclusion criteria, being both hearing and not knowing any DGS, resulting in a sample of 129 participants. The actual sample size is lower for some items due to participants choosing not to respond.

54.26 % of participants identified as female, 27.13 % as male, 10.08 % as diverse, and four chose not to answer. Only one respondent was younger than 18, 19.38 % were between the ages of 18 and 30, 27.91 % between 31 and 40, 46.51 % between 41 and 60, and 3.88 % were older than 60. 2 people chose not to disclose their age.

Most participants (81.40 %) are d/Deaf, some h/Hard of h/Hearing (11.63 %), and few (3.10 %) late-deafened. A small sub-sample of hearing participants (3.88 %), two of whom indicated DGS as their native language and three of whom possess moderate DGS skills, was included in this research. 21.70 % of participants stated that they need a translation of any form of text on websites and in apps, 16.28 % require a translation for texts comprising one or more full sentences, 33.34 % need translations only for longer text passages and complicated topics, and 21.71 % do not require any translation for written German. 7.00 % did not indicate their individual need for translation. Furthermore, the sample is characterized by a fairly high affinity to technology ($M = 3.03, SD = .52$).

3.3 Evaluation

It is reasonable to assume that users of sign language exhibit different priorities for technical and non-technical features of an avatar-based translation system depending on their proficiency in written and sign language, seeing that such features often contain textual information and might therefore differ in their understandability depending on written language proficiency. The most critical distinction to investigate is between individuals who do not require any translations from written German to DGS and individuals who need translations at least for longer sentences and/or more complicated topics. The second group represents the primary target audience for an avatar-based translation system, whereas the former

can be considered as a secondary target group. Potential differences in requirements may affect the prioritization and implementation of certain features. Therefore, we dichotomized the variable "need for translation", resulting in a split between those needing some translation (92 participants) and others needing no translation of written text (28 participants) [12]. To account for the complexities of self-assessment in language proficiency among SL users, we performed an analogous sample split based on reported DGS and Spoken German skills, comparing native speakers (107 for DGS, 41 for spoken German) to less proficient participants (21 for DGS, 84 for spoken German). Likewise, age differences in the interaction with technical systems, including avatar-based ones have been demonstrated [10]. Thus, we also employed a dichotomous sample split along the generational border between Millennials and Gen X. Moreover, this approach allows the inclusion of the participants in the youngest and oldest age groups, the individual size of which would not suffice for statistical analyses. The younger subsample of people aged 40 and lower consists of 63 participants, while the 65 participants over the age of 40 comprise the similarly sized older subsample.

We first performed a series of ANOVAs on the overall sample to check for differences in the prioritization of contents for translation and avatar features. For significant findings, post hoc tests were done accordingly. Employing the sample splits described above, we then conducted a series of t-tests to investigate group differences based on age and written language skills. Participants who either did not provide their age or did not indicate their level of comfort with written text, were not included in the respective group comparisons.

4 RESULTS

4.1 Translations of different contents

4.1.1 Importance of translations in general. Participants rated the importance of the translation of different type of contents. There is a significant difference in the level of priority ($F(3.70, 447.68) = 52.7, p < .001, \eta_p^2 = .304$). Notably, Bonferroni-adjusted post hoc comparisons revealed that the translations of announcement texts ($M = 3.58, SD = .83$) and news ($M = 3.65, SD = .72$) are significantly prioritized over other types of content, while the translation of instructions for using the site ($M = 2.72, SD = 1.05$) and explanations for forms ($M = 2.75, SD = 1.06$) are generally deemed less important. This shows that in the context of travelling, SL users are especially interested in translations of contents that require a quick reaction and less interested in information that they are able to gather at a later time or through trial and error.

4.1.2 Importance of translations for different groups. As is to be expected, persons who have a need for translations into SL at least for long sentences rate the translation of search results ($t(117) = -2.40, p = .018, d = -.520$), hints on forms ($t(116) = -2.52, p = .013, d = -.553$), content of the application ($t(117) = -2.95, p = .004, d = -.645$) and current content ($t(117) = -2.29, p = .024, d = -.495$) as more important than persons with no need for translations into SL. Furthermore, a difference in the assessment of the translation of operation instructions can be seen. These are rated as more important by older people than by the younger participants ($t(121) = 2.07, p = .040, d = .374$).

4.2 Functions of the avatar

4.2.1 Importance of avatar functions in general. There is a significant difference in the assessed importance of the avatar functions ($F(8.29, 792.62) = 17.1, p < .001, \eta_p^2 = .151$). Strikingly, a possible feature allowing users to choose or create their own avatar was ranked significantly lower than most other functions ($M = 2.62, SD = 1.02$). The highest priority features were directly related to facilitating sign recognition and understanding, including changing the avatar's size ($M = 3.48, SD = .71$), displaying additional subtitles ($M = 3.54, SD = 0.66$), adjusting the speed ($M = 3.32, SD = .87$) and marking individual words in order to look them up in a DGS dictionary ($M = 3.39, SD = .82$).

4.2.2 Importance of avatar functions for different groups. Participants who indicate a need for translations into DGS rate the importance of being able to change the size of a sign language avatar higher than person who do not indicate this need ($t(112) = 2.88, p = .005, d = .740$). Similarly, the ability to set the contrast to the background in a sign language avatar is more important for the group of persons with a need for translations into DGS than by persons without this need ($t(113) = 2.20, p = .030, d = .564$).

4.3 Appearance of the avatar

4.3.1 Importance of avatar appearance in general. The rated importance differs significantly for the options to adjust the look of the avatar ($F(4.52, 492.42) = 29.0, p < .001, \eta_p^2 = .210$). The two features that affect the recognizability of the signs the most, namely the type (level of anthropomorphism) ($M = 2.79, SD = 1.11$) and background ($M = 3.22, SD = .91$), are considered to be significantly more important than the rest.

4.3.2 Importance of avatar appearance for different groups. No significant differences for any settings related to the avatar's appearance were detected based on age and language proficiency.

5 DISCUSSION

In conducting our quantitative online survey with native DGS users, we were able to address the topic of adjustability and content selection in the context of avatar-based sign language translation technology with and for DGS users. As such, our findings constitute insights and knowledge on a diverse research agenda within this field. Contributing from a methodological perspective, our approach of an ability-oriented, DGS-accessible online survey allowed us to assess and analyse respective requirements with a large group of native DGS signers as participants. Contentwise, our results emphasize the relevance of considering adjustable parameters of sign language avatars just as well as that of considering options for prioritization in content selection when offering DGS translations.

5.1 Study design with and for DHH communities

With previous research being largely focused on evaluating SL avatars through the use of focus groups [13, 14, 18, 21, 22], best practices for designing studies with and for DHH communities concentrate on evaluation procedures with (small) groups of people, with only few studies examining the question of how to collect broader feedback from a large amount of target group members in

a quantitative manner [3, 14]. Our design of a DGS-accessible online survey thus contributes to and complements guidelines on using online surveys as a tool for research within DHH communities. In detail, our high number of participants shows that both, our DGS-accessible design of the questionnaire, as well as our community-based approach for its distribution, can be seen as promising for high response rates⁴. In addition, our study design revealed that our example-based self-assessment of the participants language background allowed us a more sensitive analysis of results. While the more standard mapping of language skills in (spoken) German ('mother tongue', 'basic knowledge', or 'no knowledge') showed little to no tendencies or variances in the results between different skill groups, the mapping of example-based language skills ('I need DGS translations at least for longer sentences or complex topics' vs. 'I don't need DGS translations') offered a stronger distinction between the answers of the groups, emphasizing for instance a demand for translation of more content types from people with a stronger self-assessed need for DGS translations. Although this distinction may seem little innovative in its design implications, it does show the importance of accurate assessment methods for the language background of participants.

5.2 Content selection for translation within travelling contexts

Although a complete translation of any information into DGS is ultimately demanded by its users⁵, our results show that there are content types within travelling contexts that require direct translation options more urgently than others. This prioritization is mainly determined by the dimension of time-pressure within specific contexts, i.e., information within high time-pressure contexts require options for translation more urgently than others. Besides time-pressure, we also identified the participants language background as an influential factor on content prioritization, indicating that people who generally require more translation also view more content types as relevant for translation than others do. As a third influential factor, our results also point towards age, indicating that older participants demand translation of operating instructions (e.g., in apps or forms) more often than younger participants. We assume this to be related to a general difference between exploration behavior when interacting with technology which is often stifled for older users by anxiety [5, 11, 31], but nevertheless contributes a valuable perspective on content selection for SL translation technology. Taken together, these three factors emphasize the importance of offering choices to users on which information they wish to translate and which not.

⁴It should be emphasized that this potential of a study design must always be considered in combination with the respective topic that the study addresses. The statements collected from our participants indicate a strong and urgent interest in the development of (improved) SL translation systems which is very likely to have positively impacted response rates.

⁵We identify this demand especially based on the optional feedback and comments provided at the end of the questionnaire. Participants used this box to request, for instance, translations for IOS and Android devices, live translations, or a 24/7 availability of the avatar.

5.3 Interface design of animation-based SL translation systems

Finally, we can deduce a general necessity of making sign language avatars more adjustable to users. Currently, this necessity relates mainly to the adjustment of parameters influencing the understandability of the avatar, i.e., its size, the display of subtitles or the design of its background. Parameters concerning subjective design choices, such as the outer appearance of the avatar, are considered to be of lesser importance for individual adjustments by SL users. It can be discussed whether this identified prioritization might be caused by the current state of respective research. With SL avatars being only in the process of initial development, users might be more focused on elementary parameters required for achieving understandable SL performances, instead of what feels like supplemental add-ons, such as the look or clothing of the avatar. As a single exception from this, participants rated it important to be able to set the avatar design to a human-like appearance, because they felt that comic-like designs may often lead to SL users feeling not taken seriously and should therefore be avoided. Overall, we understand this insight into which parameters users would like to individually adjust or configure as part of the main contribution of this paper. It emphasizes the need for future studies to design and evaluate not only the avatar in itself and its SL quality, but also usability and user experience dimensions of interfaces surrounding such avatars.

5.4 Limitations

The representativity of our study sample is limited in terms of age and nationality distribution: both very young and older people are underrepresented in our sample and the study considers only members of the German DHH community. Also, the idea of collecting attitudes about a system that does not exist yet causes an inherent decline in reliability due to participants' incorrect mental model of the system and inadequate assessments of one's future behavior. In addition to these aspects, we identified three limitations that refer to more complex challenges. They arise from the ambition to create a fully DGS-accessible quantitative online study and, as such, emphasize the mismatch of current research tools for conducting quantitative online studies within DHH communities [33].

Non-standardized item translation. Item wordings can be expected to differ slightly when translated from written German into DGS due to the two being separate languages with different linguistic concepts and elements. This implies that participants end up replying to slightly different question phrasings depending on whether they rely on the video translation, the written text, or a mix of both. In addition, intricacies in item wordings are likely to get lost in translation. With no respective standardized DGS-questionnaires available, we met this issue by ensuring professional translation standards as well as conducting pre-tests with bilingual DGS-experts. This approach does not solve the issue in itself, but promised high potential for content equality between the items and their translations.

Integrating open-ended questions in a DGS-accessible online survey. Even though our questionnaire was set as a quantitative study, qualitative elements in the form of open-ended questions might

have provided more diversity in responses or highlighted divergent views. However, including a larger amount of such open questions – while ensuring our ability-oriented approach – would have required the option of answering questions in DGS and not only in written form. Such an option brings along a variety of challenges with most of them leading to very high interpretation costs.

Self-assessment of spoken German language skills. Considering our item wording for the self-assessment of spoken German language skills, we recognize the difficulty of choosing the term 'mother tongue' as one skill level, seeing that participants might choose not to describe their skill level as such – not because they do not have the actual skills of a native speaker, but because they identify with DGS as their mother tongue and not spoken German. As noted in previous sections, we instead encourage the use of example-based item wordings for assessing the participants language background and avoid typical phrasings such as 'mother tongue' or 'native speaker'.

6 FUTURE WORK

To address the limitation of this study's purely attitudinal approach, an empirical validation is required, in which the effect of an SL translation system's characteristics and technical features on user acceptance is investigated. Our research has demonstrated the importance of example-based self-assessment of written language skills rather than relying on established language proficiency levels. We recommend that this approach is applied in future studies with SL users. Finally, we see great value in conducting similar research with users of other sign languages in order to facilitate a consistent user experience within each platform across nationalities.

7 CONCLUSION

The main contributions of the study presented in this paper accumulate to the ability-oriented and DGS-accessible design of a quantitative online questionnaire, our high number of participants and, with that, the broad amount of perspectives and feedback on the issues of adaptation / configuration and content selection within the interfaces of avatar-based DGS translation technology. This paper extends the focus of current research practices from evaluating solely the avatar and its SL quality onto the surrounding interface and its measures of usability and user experience.

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